

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning, Data integration, Data transformation and data Reduction ,for effective data preparation (BL3,BL4)

QUESTIONS: 5

Total Marks:20

Annexure A

CLASS TEST

<i>Criteria</i>	Understanding of Task (2)	Procedure & Execution (2.5)	Observation & Result (2)	Viva Voce / Conceptual Response (2)	Time Management & Presentation (1.5)	<i>Total(10)</i>
<i>Weightage</i>	20%	25%	20%	20%	15%	<i>100%</i>
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I am B V Chaitanya Reddy-192425376(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Chaitanya

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Task	20%	Clearly understands the given task/experiment without assistance	Understands task with minor clarification	Partial understanding ; requires guidance	Unable to understand the task
Procedure & Execution	25%	Performs procedure accurately and independently with proper technique	Performs with minor errors	Requires guidance; makes procedural mistakes	Unable to execute the procedure
Observation & Result	20%	Records accurate observations and obtains correct results	Minor errors in observation or result	Incomplete or partially correct results	Incorrect or no results
Viva Voce / Conceptual Response	20%	Answers all questions confidently with strong conceptual clarity	Answers most questions with minor gaps	Limited responses; needs prompting	Unable to answer questions
Time Management & Presentation	15%	Completes task on time with neat and clear presentation	Completes with minor delay or presentation issues	Delayed or poorly presented work	Unable to complete on time

CLASS TEST

QUESTIONS:

Total Marks: 10x2 =20

Q1.

- a) Define Data Mining and list any two basic data mining tasks. Explain the steps involved in KDD process and justify the role of data preprocessing in it. [L2, L4]
- b) What is data cleaning? Mention two common data quality issues. Explain methods to handle missing values and noisy data with suitable examples. [L1, L2, L3]

Q2.

- a) Define data integration and list two challenges associated with it. Explain how redundancy and inconsistency are handled during data integration. [L2, L4]
- b). What is covariance analysis? List the two covariance formalms[L2, L3]

Q3.

- a) Suppose two stocks A and B have the following values in one week: (2, 5), (3, 8), (5, 10), (4, 11), (6, 14). Question: If the stocks are affected by the same industry trends, will their prices rise or fall together and independent ?
- b) Explain the architecture of a typical data mining system with a neat diagram. Discuss the role of each component and analyze how they interact to support knowledge discovery. [L2, L4]

Q4..

- a) Find covariance for following data set $x = \{4,5,6,7,9\}$, $y = \{8,3,7,5,6\}$
- b) Describe the major steps in data pre-processing. Given a dataset with missing values, outliers, and inconsistent formats, propose a systematic pre-processing pipeline and justify the sequence of steps. [L3, L4, L5]

Q5

a). A hospital is analyzing patient **age** data to identify broad health trends. Because the raw data contains minor variations and noise, the analyst needs to perform local smoothing using **equal-frequency** (equi-depth) binning. **Sample Data Set (Ages):** 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70 [L2, L4]

b). Explain data discretization and its importance in data mining. Compare equal-width and equal-frequency binning with suitable examples, and evaluate their impact on data analysis. [L3, L4, 15]

CO2 - AT2

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CSA1613

① a) Data mining and KDD :-

Data mining :- process of finding useful patterns from large data.

Tasks :-

01) Classification.

02) Clustering.

KDD steps :-

- Data cleaning.
- Data integration.
- Data selection.
- Data transformation.
- Data mining.
- Pattern evaluation.
- Knowledge presentation.

Role of preprocessing :-

- Improves the data quality, removes missing values, noise and increases mining accuracy.

89 / 100

1) b) data cleaning :-

- correcting incomplete and the incorrect data.

Issues :-

- Missing value.
- noisy data.

Missing value :- Replacing using the mean, median and mode.

Ex: 20, 25, —, 30 → replace with 25.

Noisy data :-

- use binning or smoothing.

② a) Data integration :-

- Combining data from different sources into one dataset.

Challenges :-

- Redundancy and inconsistency.

Redundancy :-

- Remove duplicate records.

Inconsistency :-

- convert data into a common format
for example: data format.

b) Architecture of a data mining System:-

Typical data mining system architecture;

Data sources / data bases



data cleaning and integration



Data warehouse / repository



Data mining engine



pattern evaluation



user interface / knowledge

presentation.

Components:-

Data sources, pre processing, warehouse, mining engine, evaluation and user interface.

2) Covariance Analysis :-

- Covariance measures the relationship between two variables.

Formulae:

for population:

$$\text{COV}(X, Y) = \frac{\sum (x - \bar{x})(y - \bar{y})}{N}$$

for sample:

$$\text{COV}(X, Y) = \frac{\sum (x - \bar{x})(y - \bar{y})}{n-1}$$

- positive covariance means variables increase together, negative covariance means they move oppositely.

3) a) Stock Covariance:

Given

(2, 5), (3, 8), (5, 10), (4, 11), (6, 14)

- As stock A increases, stock B also generally increases.

∴ covariance positive

• The stock prices rise together and are positively related, not independent.

a) Covariance :

Given data

$$X = (4, 5, 6, 7, 9)$$

$$Y = (8, 3, 7, 5, 6)$$

$$\text{Mean}(X) = \frac{4+5+6+7+9}{5} = 6.2$$

$$\text{Mean}(Y) = \frac{8+3+7+5+6}{5} = 5.8$$

$$\text{Covariance} = -0.36$$

Interpretation : slight negative relationship.

b) Data preprocessing pipeline :

Steps :-

- Handle missing values
- Remove duplicates
- Correct formats
- remove outliers
- Normalization
- Reduce data.

• This order improves the data quality and accuracy.

5) a) Equal frequency binning:

given data

13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70.

total values = 27

Bins

B1 = 13, 15, 16, 16, 19, 20, 20, 21, 22 mean = 18

B2 = 22, 25, 25, 25, 30, 33, 33, 35 mean = 28.11

B3 = 35, 35, 35, 36, 40, 45, 46, 52, 70 mean = 43.78

- equal-frequency binning smooths the age data and reduces noise.

5) b) Data discretization:-

- Data discretization converts continuous values into intervals or bins.

Importance?

Reduces complexity and improves mining

~~equal-width~~

~~equal-frequency.~~

- equal interval size

- equal no. of values

- simple

- Better for skewed data.